

Skylark Business Intelligence Agent

A conversational agent that answers founder-level business questions by querying two monday.com boards live — sales pipeline (**Deals**) and project delivery (**Work Orders**).

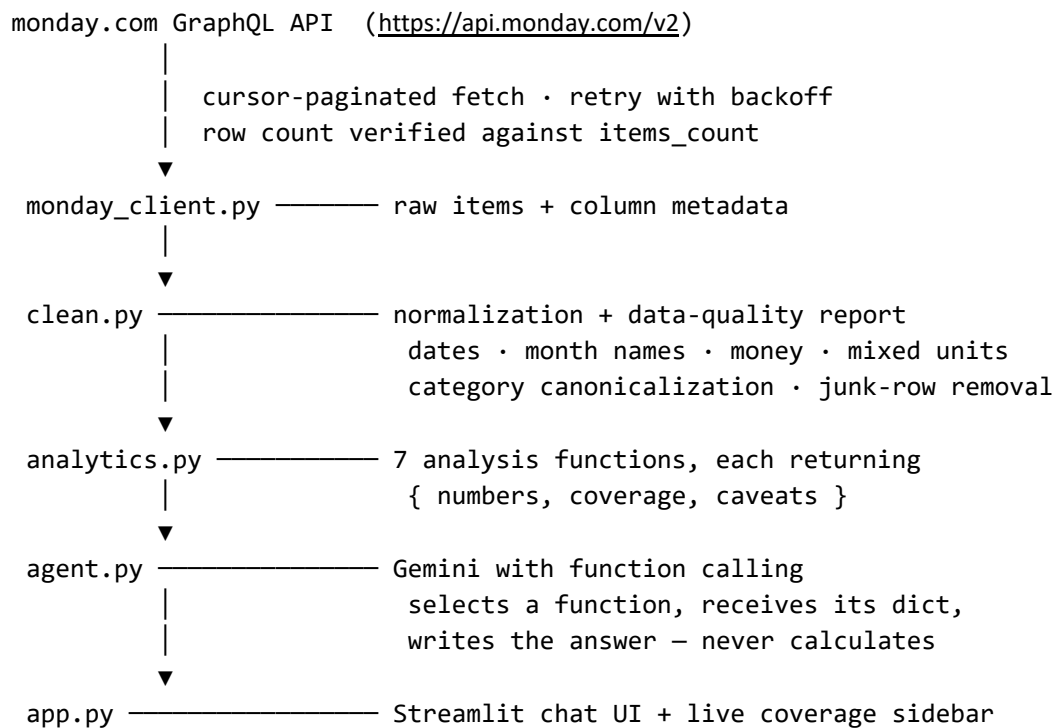
Live prototype: [<https://skylark-bi-agent-aoqx2me7kbvpoapptdjtpr.streamlit.app/>]

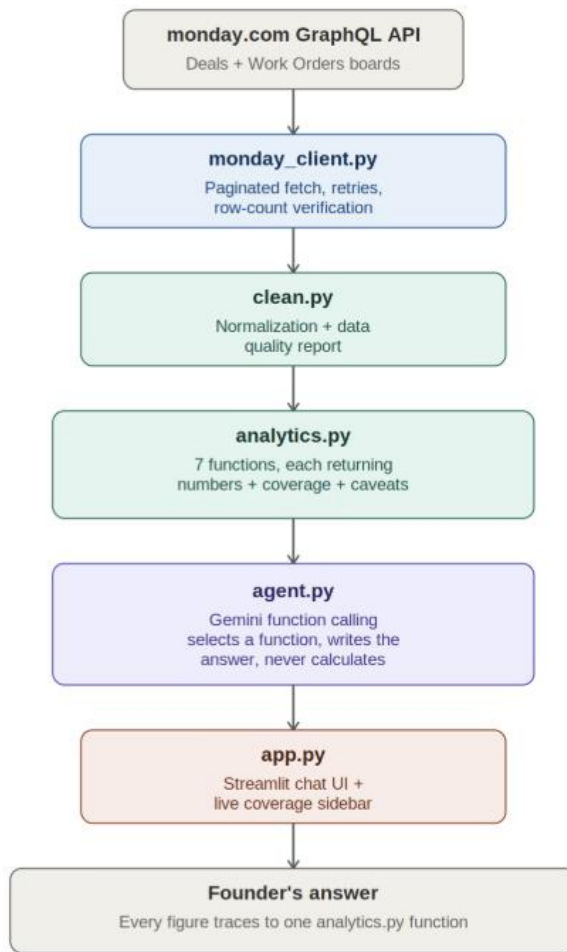
"How's our pipeline looking for the energy sector this quarter?" "Which projects are running late?" · "Give me a leadership update."

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Architecture





The model never touches raw data and never does arithmetic. It chooses which analysis function to run and explains the result. Every figure in every answer traces back to a function in `analytics.py`, which makes fabricated numbers structurally impossible.

Files

File	Role
<code>monday_client.py</code>	GraphQL client. Pagination, retries, per-board error isolation.
<code>clean.py</code>	API response → tidy DataFrame. All normalization and quality reporting.
<code>analytics.py</code>	Business logic. Seven functions, each with coverage and caveats.
<code>agent.py</code>	Gemini tool-calling loop, model fallback, rate-limit handling.
<code>app.py</code>	Streamlit interface and design system.
<code>quota_check.py</code>	Diagnostic: which model works, which quota is exhausted.

The seven analysis functions

Function	Answers
<code>pipeline_summary</code>	Open deals by stage and probability, with value coverage
<code>win_rate</code>	Won vs Dead, with mean/median deal size and outlier detection
<code>deals_by_sector</code>	Per-sector counts, values, win rates, concentration warnings

revenue_summary	Order book, billing, collections, receivables (GST-consistent)
operational_health	Execution status, overdue work orders, recurring contracts
sector_performance	Cross-board: sales next to delivery, per sector
leadership_brief	Everything above, bundled, plus known blind spots

monday.com setup

1. Create two boards

Import each provided spreadsheet as its own board via **Add → Import data → Excel/CSV**.

Board	Rows
Work Orders	176
Deals	344

Column types can be left largely to monday's own detection. Two worth correcting:

- **Serial # → Text**, not Dropdown. It holds 176 unique values.
- **Month-name columns → Text**, not Date (Last executed month, Actual Billing Month). They contain bare month names with no year.

Import the data **as-is**. Do not clean it in the spreadsheet — handling the mess at runtime is the point, and `clean.py` covers every defect present in the sample data.

2. Get an API token

Avatar (top right) → **Administration → Connections → Personal API token** → Copy. Non-admins: avatar → **Developers → My Access Tokens**.

3. Get the board IDs

Open each board. The long number in the browser URL is its ID: <https://your-team.monday.com/boards/5030963827>

4. Get a Gemini API key

aistudio.google.com/apikey → **Create API key**. Free tier, no card required.

If you hit a daily quota, create a key in a **new project** — quota is allocated per project, so a new project gets a fresh allowance immediately.

Running locally

Requirements: Python 3.10+

bash

```
pip install -r requirements.txt
```

Create a `.env` file in the project root:

```
MONDAY_TOKEN=your_monday_personal_api_token
WORK_ORDERS_BOARD_ID=1234567890
DEALS_BOARD_ID=1234567891
GEMINI_API_KEY=your_gemini_key
GEMINI_MODEL=gemini-3.5-flash-lite
```

Verify the monday connection before starting the UI:

```
bash
python monday_client.py
```

Expected — the counts must match your boards:

```
Work Orders: 176 items, 38 columns
              counts OK
Deals: 344 items, 12 columns
              counts OK
```

Then launch:

```
bash
python -m streamlit run app.py
```

Deploying

Push to a **public** GitHub repository, then deploy on share.streamlit.io with `app.py` as the entry point.

Under **Advanced settings** → **Secrets**, add the same values in **TOML** format — note the quotes and the spaces around `=`, which differ from `.env`:

```
toml
MONDAY_TOKEN = "... "
WORK_ORDERS_BOARD_ID = "5030963827"
DEALS_BOARD_ID = "5030963877"
GEMINI_API_KEY = "... "
GEMINI_MODEL = "gemini-3.5-flash-lite"
```

`.env` is gitignored and must never be committed.

Configuration reference

Variable	Required	Purpose
MONDAY_TOKEN	yes	monday.com personal API token
WORK_ORDERS_BOARD_ID	yes	Work Orders board ID

DEALS_BOARD_ID	yes	Deals board ID
GEMINI_API_KEY	yes	Google AI Studio key
GEMINI_MODEL	no	Defaults to gemini-3.6-flash
STATUS_WON	no	Labels meaning a won deal. Default Won
STATUS_LOST	no	Labels meaning a lost deal. Default Dead
STATUS_OPEN	no	Default Open
STATUS_ON_SCHEDULE	no	Statuses not counted as late. Default Completed, Executed until current month
SMALL_SAMPLE_THRESHOLD	no	Below this, percentages are flagged unreliable. Default 5

The STATUS_* variables exist so a monday account using different labels — say Closed Won instead of Won — needs a config change rather than a code change.

Design decisions

No hardcoded data. There is no file I/O anywhere in the codebase — no read_csv, no read_excel, no open(). Board IDs and credentials come from the environment. Removing them makes the app produce nothing, which is the intended proof.

Data is fetched once per session and cached in memory, then reused across questions to respect rate limits. **Refresh from monday.com** in the sidebar clears the cache and re-queries live.

Pagination is verified. Each board's fetched row count is checked against its reported items_count. Ignoring monday's cursor would silently return the first 100 rows and produce confident, wrong totals.

Errors degrade rather than crash. If one board fails, the other still loads. Retired models are handled by reading the replacement name out of the provider's error response. Rate limits retry with the server's suggested delay.

Coverage is a first-class output. Every number carries the share of records behind it, and the sidebar renders those as ticked survey rules that shift colour as coverage drops. Entirely empty columns are named as untracked rather than reported as zero.

See DECISION_LOG.md for assumptions, trade-offs, and the full list of data-quality issues found.

Troubleshooting

Symptom	Cause and fix
WORK_ORDERS_BOARD_ID and DEALS_BOARD_ID must both be set	.env not loaded, or saved as .env.txt on Windows. On Streamlit Cloud, secrets not saved.
monday.com rejected the token (401)	Token regenerated or mis-copied. Get a fresh one from Administration → Connections.
404 NOT_FOUND on the model	Model retired. The app reads the replacement from the error and retries automatically; set GEMINI_MODEL to skip the failed attempt.

429 / rate limit message	Free-tier quota. Per-minute clears itself. For per-day, create a key in a new project. Run <code>python quota_check.py</code> to tell which.
503 UNAVAILABLE	Provider-side spike. Retried automatically, then falls back to another model.
Row counts don't match the board	Pagination or permissions. Run <code>python monday_client.py</code> to isolate.
streamlit: command not found	Use <code>python -m streamlit run app.py</code> .